

A (Hopefully!) Intuitive Guide to Linear Algebra

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An introduction: why linear algebra?

Linear algebra is the language of vectors, matrices, and linear transformations. It is useful whenever many quantities move together: in statistics and econometrics, in optimization, in machine learning, in numerical analysis, and throughout applied mathematics. Although matrices take some time to get used to, their main purpose is to simplify our thinking and reveal the structure of the underlying math more clearly. More specifically,

- **They keep the algebra under control.** A regression with one explanatory variable and a regression with a thousand explanatory variables have the same matrix formula. Once the dimensions are clear, a single proof handles both cases.
- **They reveal geometry.** A matrix can be viewed as a transformation of space. Least squares then becomes a projection; rank records how many independent directions survive a transformation; and a determinant measures how oriented volume changes. Put another way, matrices give us an additional source of intuition that can help us get our heads around the more difficult concepts in econometrics.

These notes assume that you have seen algebra and basic calculus before, but they do not assume that either is fresh in your mind. Before we start, I have a few small pieces of practical advice. One general principle is worth keeping nearby:

More abstract math usually comes from generalizing a familiar idea.

For example, a scalar can be represented as a 1×1 matrix, and a vector can be represented as a one-column matrix. When a matrix identity feels unfamiliar, it's often useful to ask what it would mean first for ordinary numbers. You can then add the extra 'linear algebra' elements on top, such as the extra care required because $AB \neq BA$ in general for matrices A, B . I've always found this a very useful way to think about all kinds of math, and I'd recommend you to try to organize your thinking like this where you can. In addition, if you have any problem with understanding any concept in later courses, there is a good chance that the problem may be with the fundamental mathematical concepts, rather than the new things you're learning. Feel free to

return to these notes whenever helpful until these things feel intuitive. Finally, if you find yourself getting lost when writing your own proofs, it's often because a small piece of the underlying math has been used silently: maybe a dimension has changed, an inverse has been assumed to exist, or the order of a product has been reversed. Writing dimensions beside the objects is very helpful, and I'd always recommend that you do it.

The symbol $\mathbb{R}$ denotes the set of real numbers. A *scalar* is a single number in $\mathbb{R}$; $\mathbb{R}^n$ is the set of real n -component column vectors; and $\mathbb{R}^{m \times n}$ is the set of real $m \times n$ matrices. Unless stated otherwise, all scalars, vectors, and matrices in these notes take values in the real (rather than complex) numbers. Vectors are columns, and a prime denotes transpose: x' is the row vector obtained by transposing x . The identity matrix of size n is I_n , and a bold zero denotes a zero vector or matrix of the dimension required by context. For $x \in \mathbb{R}^n$, the Euclidean norm (or length) is $\|x\| = \sqrt{x'x}$.

The basics: what is a matrix, and why do we care?

The key to understanding a matrix is to become comfortable moving between two ways of viewing it. A matrix is **both**:

- **A table of entries.** If A has m rows and n columns, we write $A \in \mathbb{R}^{m \times n}$, or describe it as ' m by n '. Its entry in row i and column j is denoted a_{ij} , A_{ij} or $[A]_{ij}$ (depending on the author). A matrix is *square* if $m = n$ and *rectangular* if $m \neq n$.
- **A linear transformation.** The same $A \in \mathbb{R}^{m \times n}$ defines a function from $\mathbb{R}^n$ to $\mathbb{R}^m$ by sending x to Ax . A 'transformation' is just another word for a function.

For example, if

$$A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \quad \text{and} \quad x = \begin{bmatrix} 4 \\ -2 \end{bmatrix} \quad \text{then} \quad Ax = \begin{bmatrix} 2(4) + 1(-2) \\ -1(4) + 3(-2) \end{bmatrix} = \begin{bmatrix} 6 \\ -10 \end{bmatrix}.$$

The reason why this is so fundamental is that some concepts in linear algebra are much easier to understand if we take the table view, and some are much easier if we take the function view. For example, the concept of a trace is very easy to understand if we picture a matrix as a table in our minds, but its geometric interpretation is much more difficult. On the other hand, it is much easier to view a determinant in terms of what the transformation does to the area of the unit square, rather than its formula. Below, we take whichever view makes each concept easier to understand, while keeping in mind that being able to take both views is an important part of feeling comfortable with matrices. The simplest properties of matrices are just the same as they are for vectors. Matrices of the same shape are added entry by entry, and multiplying a matrix by a scalar multiplies every entry by that scalar. These simple rules make linear combinations of matrices behave just like linear combinations of vectors. An expression of the form $a_1v_1 + \cdots + a_rv_r$, where the a_i are scalars, is called a *linear combination* of $v_1, \dots, v_r$.

A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear if, for every $u, v \in \mathbb{R}^n$ and scalars $a, b \in \mathbb{R}$,

$$T(au + bv) = aT(u) + bT(v).$$

Here, $\mathbb{R}^n$ is the *domain* and $\mathbb{R}^m$ is the *codomain*. The *image*, or *range*, of T is the set $\{T(x) : x \in \mathbb{R}^n\}$. The function is *one-to-one* if different inputs give different outputs, *onto* if its image equals its codomain, and *bijective* if it is both one-to-one and onto. In particular, $T(\mathbf{0}) = \mathbf{0}$, where $\mathbf{0}$ is a vector consisting only of zeros. The function $T(x) := Ax$ for any matrix A is always linear, which is why we say above that matrices represent linear transformations. By contrast, if $c \in \mathbb{R}^n$ is non-zero, $T(x) = x + c$ is an *affine* rather than a linear transformation because $T(\mathbf{0}) = c \neq \mathbf{0}$. Linearity underlies many of the key properties of matrices.

Furthermore, let $e_1, \dots, e_n$ be the standard coordinate vectors: e_j has a 1 in position j and zeros elsewhere. The j th column of A is exactly Ae_j . Since every $x \in \mathbb{R}^n$ can be written as $x = x_1e_1 + \cdots + x_n e_n$, linearity gives

$$Ax = x_1 Ae_1 + \cdots + x_n Ae_n.$$

As such, we can view Ax as the vector made up by linearly combining the columns of A - $Ae_1, \dots, Ae_n$ - by the (scalar) components of x - $x_1, \dots, x_n$. For the matrix above,

$$Ax = x_1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

This single observation will explain matrix multiplication, column spaces, rank, determinants, change of basis, and least squares.

Different ways to view a data-generating process

Data often arrive in the form of a table, with observations arranged in rows and variables arranged in columns, and this table can be viewed naturally as a matrix. A *data-generating process* (DGP) is a description of how observable variables and unobserved terms arise together. Matrix notation does not alter this process, nor does it introduce new assumptions, but it gives us a new notation in which to view and work with the DGP. Let i index one of n observations, let y_i be an outcome and z_i an explanatory variable, let β_0 and β_1 be fixed coefficients, and let ε_i be an unobserved error. The usual **scalar** way of writing a DGP is

$$y_i = \beta_0 + \beta_1 z_i + \varepsilon_i, \quad \mathbb{E}(\varepsilon_i \mid z_i) = 0. \quad (1)$$

To write the same model in vector form, define

$$x_i = \begin{bmatrix} 1 \\ z_i \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}.$$

Note that

$$x_i' \beta = [1 \quad z_i] \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \beta_0 + \beta_1 z_i,$$

so the model can be written in **vector** form as:

$$y_i = x_i' \beta + \varepsilon_i, \quad \mathbb{E}(\varepsilon_i \mid x_i) = 0. \quad (2)$$

Here, the prime denotes a transpose. The two conditional-mean statements are equivalent because x_i contains exactly the same information as z_i ; if you know what value x_i takes, then you know what value z_i takes, and vice versa. The main advantage of this vector notation is that its form remains unchanged when further explanatory variables are introduced. If there are k regressors (including the constant), then $x_i, \beta \in \mathbb{R}^k$, and the DGP is still written using the notation $x_i' \beta$.

To get things into matrix form, we now combine, or *stack*, the equations for all n observations. Before, we wrote everything with an i subscript, denoting the fact that this equation holds for every i . To make this into one expression, we define

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n, \quad X = \begin{bmatrix} 1 & z_1 \\ 1 & z_2 \\ \vdots & \vdots \\ 1 & z_n \end{bmatrix} \in \mathbb{R}^{n \times 2}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \in \mathbb{R}^2, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix} \in \mathbb{R}^n.$$

Note that each row of X contains the regressor values for one observation, and each column contains the values of one regressor across all observations. More generally, with k regressors in total, counting the intercept, $X \in \mathbb{R}^{n \times k}$. The equations can therefore be written in **matrix** form as:

$$y = X\beta + \varepsilon. \quad (3)$$

The dimensions provide a useful check that the matrix operations are well defined:

$$\underbrace{y}_{n \times 1} = \underbrace{X}_{n \times k} \underbrace{\beta}_{k \times 1} + \underbrace{\varepsilon}_{n \times 1}.$$

Since the product of an $n \times k$ matrix and a $k \times 1$ vector is an $n \times 1$ vector, every term in the equation has the same dimensions. For example, let

$$X = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix}, \quad \beta = \begin{bmatrix} 3 \\ 1.5 \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} 0.2 \\ -0.1 \\ 0.4 \end{bmatrix}.$$

Then

$$X\beta = \begin{bmatrix} 1.5 \\ 3 \\ 6 \end{bmatrix}, \quad y = X\beta + \varepsilon = \begin{bmatrix} 1.7 \\ 2.9 \\ 6.4 \end{bmatrix}.$$

Thus each row reproduces one scalar equation; for instance, the second row gives $2.9 = 3 + 1.5(0) - 0.1$.

The reason why we have three different notations for the DGP is that they are useful in different circumstances.

- The **scalar** form is the best way to think about which regression you want to run, and so the interpretation of the regression itself. This is the one you see in economic papers and in applied economics more generally.
- The **vector** form is the best way to write things when proving that ordinary least squares (OLS) is *consistent*, meaning that the estimator approaches the true coefficient as the sample grows. Vector notation naturally creates terms which look like ' $\frac{1}{n} \sum_{i=1}^n (\text{something})$ ', to which we can apply the law of large numbers and the central limit theorem (CLT). The same approach applies to other methods, such as IV. As discussed above, one proof works no matter how many variables there are in the regression.
- The **matrix** form is natural when proving that OLS is *unbiased*, when studying standard errors or when studying weak instruments. Like the vector form, it applies no matter how many variables there are in the regression.

The key to these is the idea that the matrix $X'X$ can also be written as $\sum_{i=1}^n x_i x_i'$, where X is the object in the matrix form and x_i is the object in the vector form. This is something we will prove in lecture. Some econometricians will directly consider terms like $\frac{1}{n} X'X$ rather than changing notation first, but in my view doing so helps to correctly apply things like the LLN and CLT correctly. Moving between these is—in my view—one of the best ways to make econometrics proofs as simple as possible.

The basic concepts

Dimensions and transpose

If $A \in \mathbb{R}^{m \times n}$, its transpose $A' \in \mathbb{R}^{n \times m}$ is obtained by interchanging rows and columns: $(A')_{ij} = A_{ji}$. For example,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \implies A' = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

Useful rules, whenever the displayed products are defined, are

$$(A')' = A, \quad (A + B)' = A' + B', \quad (AB)' = B' A'.$$

Note that when transposing a product, we have to swap the order of the matrices. For vectors $x, y \in \mathbb{R}^n$, the product $x'y = \sum_{j=1}^n x_j y_j$ is a scalar, and is known as the *dot product*¹. Since a scalar equals its own transpose, $x'y = (x'y)' = y'x$. For column vectors $x \in \mathbb{R}^m$ and $y \in \mathbb{R}^n$, the matrix $xy' \in \mathbb{R}^{m \times n}$ is called their *outer product*. We describe two vectors as being *orthogonal* if the dot product between them is 0. This comes from the following formula: for vectors x and y with angle θ (in radians) between them,

$$x'y = \|x\| \|y\| \cos \theta.$$

More generally, the word orthogonal refers to any time when an inner product is 0. This is the reason why you may have seen the word orthogonal used to mean 'uncorrelated' or even 'unrelated' in the past.

¹This is a special case of a more general concept called an *inner product*. Some people will directly call this an inner product instead.

Multiplying matrices

If $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, their product $AB \in \mathbb{R}^{m \times p}$ is defined by

$$(AB)_{ij} = \sum_{\ell=1}^n a_{i\ell} b_{\ell j}.$$

The inner dimensions must match. One way to remember the rule is

$$(m \times n)(n \times p) \longrightarrow (m \times p),$$

so matrices can be multiplied together if their *inner dimensions* match, and the resulting product has the dimensions given by the *outer numbers*. For example,

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 2 & -2 \end{bmatrix}.$$

Remember that vectors - by which I mean *column vectors* - are $n \times 1$ matrices. As such:

- A matrix multiplied by a matrix gives a matrix
- A matrix multiplied by a vector gives a vector

Don't simply learn these: convince yourself they are true by writing down dimensions and checking using the expression above. Even if you don't keep track of dimensions all the time, asking yourself whether or not an object is a scalar, a vector or a matrix is usually very helpful to do to check that your dimensions are right. For example, if I take a column vector x and a matrix A , it's worth convincing yourself that the *quadratic form* $x'Ax$ is a scalar.

Matrix multiplication is associative, meaning that you move parentheses as so $(AB)C = A(BC)$. This ultimately holds because both sides map $x \mapsto ABCx$ and therefore represent the function that applies C , then B , then A . Importantly, matrix multiplication is usually *not* commutative: AB and BA may have different values, different meanings, or even different dimensions. For an explicit example, let

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Here, applying R to a vector rotates it through 90 degrees, and applying S stretches its first coordinate. Note that

$$RS = \begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix} \neq SR = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}.$$

This makes sense, as stretching along an axis and then rotating is not the same as rotating and then stretching.

Linear transformations and grid lines

Let c_1 and c_2 be fixed vectors with $c_2 \neq \mathbf{0}$, and let t be a scalar. A line can be written as $c_1 + tc_2$, where c_1 is a point on the line and c_2 gives the direction of the line. A linear transformation sends this line to $A(c_1 + tc_2) = Ac_1 + tAc_2$. This follows almost immediately from the definition of a linear transformation, but it has an important meaning: a linear transformation maps a straight line to another straight line, or to a single point if $Ac_2 = \mathbf{0}$. The expression $Ac_1 + tAc_2$ has the same form as the line $c_1 + tc_2$ that we started with, with the point Ac_1 and direction vector Ac_2 replacing c_1 and c_2 . As such, when visualizing the effect of applying a linear transformation to every point on a grid, straight lines remain straight or collapse; they cannot become curved. This explains the kinds of effects that linear transformations can produce: they can stretch, shrink, reflect, rotate, shear, collapse, or combine these effects, but they cannot bend a straight line or stretch different parts of the same line by unrelated amounts.

Figure 1 gives an example of what it looks like to multiply every point on the grid by the matrix

$$A = \begin{bmatrix} 1 & 0.6 \\ 0.25 & 1 \end{bmatrix}.$$

The standard coordinate directions are mapped to the columns $(1, 0.25)'$ and $(0.6, 1)'$, and every grid point follows by linearity.

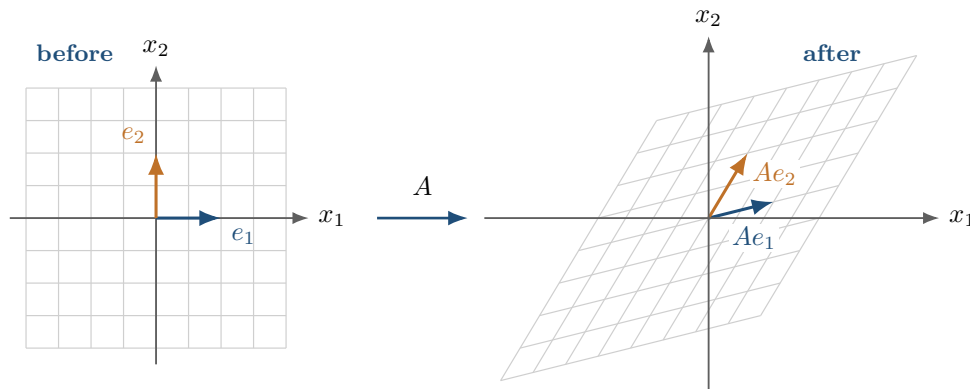


Figure 1: Grid lines under a linear transformation. Straight, parallel families remain straight and parallel, although their angles and spacing can change.

The identity and the inverse

The identity matrix I_n has ones on its diagonal and zeros elsewhere, and it leaves every vector $x \in \mathbb{R}^n$ unchanged, since $I_n x = x$. It's worth checking once in your life by explicitly multiplying out that:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix},$$

and see how the logic extends to higher dimensions. The identity matrix therefore plays the same role in matrix multiplication that the number 1 plays in ordinary multiplication. For a square matrix $A \in \mathbb{R}^{n \times n}$, an inverse is a matrix A^{-1} satisfying $A^{-1}A = AA^{-1} = I_n$. Therefore, if $y = Ax$, multiplying by A^{-1} gives $A^{-1}y = A^{-1}Ax = x$. An inverse exists exactly when the transformation represented by A is *bijective*: one-to-one and onto, so every $y \in \mathbb{R}^n$ is produced by exactly one $x \in \mathbb{R}^n$. Note that a square matrix has the same number of rows and columns, and so can be thought of as representing a transformation from $\mathbb{R}^n \rightarrow \mathbb{R}^n$. If instead $A \in \mathbb{R}^{m \times n}$ is rectangular, the transformation is from $\mathbb{R}^n \rightarrow \mathbb{R}^m$, so the domain and codomain have different dimensions. As such, the transformation cannot be bijective, and so non-square matrices are never invertible in the classic sense².

For example, if $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$, then direct multiplication gives $AA^{-1} = A^{-1}A = I_2$. More generally, for a 2×2 matrix,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \quad ad - bc \neq 0.$$

Not every square matrix is invertible: a non-invertible square matrix is called *singular*, so an invertible one is sometimes called *non-singular*. A square matrix A is invertible exactly when $Ax = \mathbf{0}$ has only the solution $x = \mathbf{0}$. To see why, suppose instead that $Ax = \mathbf{0}$ for some $x \neq \mathbf{0}$. If A^{-1} existed, then $x = A^{-1}Ax = A^{-1}\mathbf{0} = \mathbf{0}$, which would be a contradiction. If A and B are invertible, then $(AB)^{-1} = B^{-1}A^{-1}$. In fact, $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I$, and similarly $(AB)(B^{-1}A^{-1}) = I$. The order reverses because, in the product ABx , the transformation B is applied first and A second, so A must be undone before B .

²A generalized inverse for a rectangular matrix, the Moore–Penrose pseudoinverse, is introduced briefly in the least-squares section.

The unit square, the determinant, and oriented volume

In $\mathbb{R}^2$, the unit square consists of points $x_1e_1 + x_2e_2$ with $0 \leq x_1, x_2 \leq 1$. Put more simply, it is the square whose vertices are $(0, 0)$, $(0, 1)$, $(1, 0)$, and $(1, 1)$. Figure 2 applies

$$A = \begin{bmatrix} 3 & 1 \\ 1/2 & 5/2 \end{bmatrix}$$

to every point in this square. Its columns are $Ae_1 = (3, 1/2)'$ and $Ae_2 = (1, 5/2)'$. More generally, for a square matrix $A \in \mathbb{R}^{n \times n}$, the *determinant* $\det(A)$ is a scalar whose absolute value is the n -dimensional volume of the unit hypercube after A has been applied to every point. In two dimensions, this is the area of the transformed unit square. For the displayed matrix, $\det(A) = 3(5/2) - 1(1/2) = 7$, as the figure illustrates.

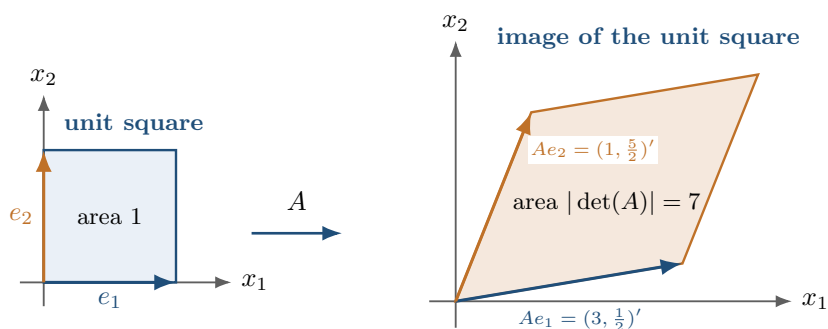


Figure 2: A matrix sends the unit square to the parallelogram whose adjacent sides are its columns.

The sign of the determinant records orientation:

- $\det(A) > 0$ means orientation is preserved;
- $\det(A) < 0$ means orientation is reversed;
- $\det(A) = 0$ means n -dimensional volume collapses to zero.

In two dimensions, orientation reversal can be pictured as a flip around one of the axes. In higher dimensions it is more precise to speak of reversing orientation rather than counting flips over particular axes. For a 2×2 matrix, the determinant has a simple formula (which we have already seen in the formula for the 2×2 matrix inverse)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \det(A) = ad - bc.$$

For instance, $A = \begin{bmatrix} 2 & 1 \\ 0 & 1.5 \end{bmatrix}$ has determinant 3: it triples area and preserves orientation. When A is invertible, its inverse scales volume by $1/|\det(A)|$ because

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

Other useful rules are

$$\det(AB) = \det(A)\det(B), \quad \det(A') = \det(A), \quad \det(cA) = c^n \det(A) \quad (A \in \mathbb{R}^{n \times n}).$$

The exponent in the last rule matters: multiplying the whole matrix by c scales each of n independent directions by c . In two dimensions, if $\det(A) = 0$, the image of the unit square has zero area. For example,

$$A = \begin{bmatrix} 1 & 2 \\ 1/2 & 1 \end{bmatrix}$$

has a second column equal to twice the first. Both coordinate directions therefore land on the same line, as Figure 3 shows. Different inputs can give the same output; for example, both $A(-2, 1)'$ and $A(0, 0)'$ are equal to the 0 vector. Once those inputs have been merged, no inverse can separate them again. As such, a matrix is invertible if and only if it has a non-zero determinant. This fits with something we have already seen: we have to assume that $ad - bc \neq 0$ for the formula for the 2×2 matrix inverse to be valid.

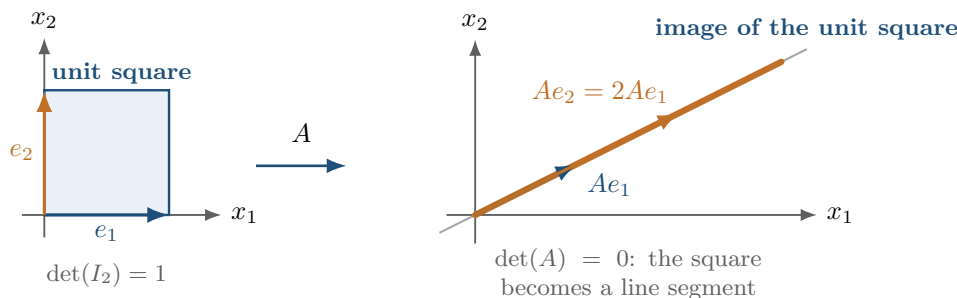


Figure 3: A singular transformation. The unit square collapses to a line segment inside the one-dimensional image of A .

Span, linear independence, and basis

A *vector space* is a non-empty set of vectors that is closed under vector addition and scalar multiplication. A subset of a vector space that has these same properties is called a *subspace*. There's quite a lot of jargon in this section, but the ultimate aim is to help us to get a rigorous definition of a *dimension* of a space. For example, we experience the world as having three spatial dimensions³: up-down, left-right and forward-back. If you've ever wondered why there are three of them, and not more or less than that, then this is the reason why.

The span of vectors $v_1, \dots, v_r \in \mathbb{R}^n$ is the set of all linear combinations that can be formed from them:

$$\text{span}\{v_1, \dots, v_r\} = \{a_1 v_1 + \dots + a_r v_r : a_1, \dots, a_r \in \mathbb{R}\}.$$

Geometrically, the span is the set of points that can be reached by moving any signed amount in the directions given by $v_1, \dots, v_r$. For example, $\text{span}\{(1, 0)'\}$ is the horizontal axis, whereas $\text{span}\{(1, 0)', (0, 1)'\} = \mathbb{R}^2$. By contrast, $\text{span}\{(1, 0)', (2, 0)'\}$ is still only the horizontal axis, because the second vector changes the available scale without adding a new direction.

The vectors $v_1, \dots, v_r$ are *linearly independent* if $a_1 v_1 + \dots + a_r v_r = \mathbf{0}$ implies $a_1 = \dots = a_r = 0$. In other words, the zero vector can be formed from them only by assigning every coefficient the value zero. This is equivalent to requiring that none of the vectors can be written as a linear combination of the others, since any non-zero combination equal to $\mathbf{0}$ reveals that at least one direction is redundant. For instance, $(1, 0)'$ and $(2, 0)'$ are linearly dependent because $2(1, 0)' - (2, 0)' = \mathbf{0}$.

A *basis* for a vector space is a linearly independent set that also spans the space. The spanning condition ensures that every vector in the space can be represented as a linear combination of the basis vectors, while linear independence ensures that this representation is unique. Finally, the *dimension* of a finite-dimensional vector space is the number of vectors in any basis for it. Every basis of $\mathbb{R}^n$ contains exactly n vectors, which is why $\mathbb{R}^n$ has dimension n ; in particular, the standard coordinate vectors $e_1, \dots, e_n$ form its standard basis. Figure 4 contrasts two independent vectors, whose linear combinations fill the plane, with two dependent vectors, whose linear combinations remain confined to a single line.

So the answer to the above question in simple: because the vectors up-down, left-right and forward-back are all linearly independent (as you can't make one of them by linearly combining the other two), and they span the space (you can reach any point by combining those three). As such, the space is $\mathbb{R}^3$.

³Leaving aside much more complicated things we know from modern physics!

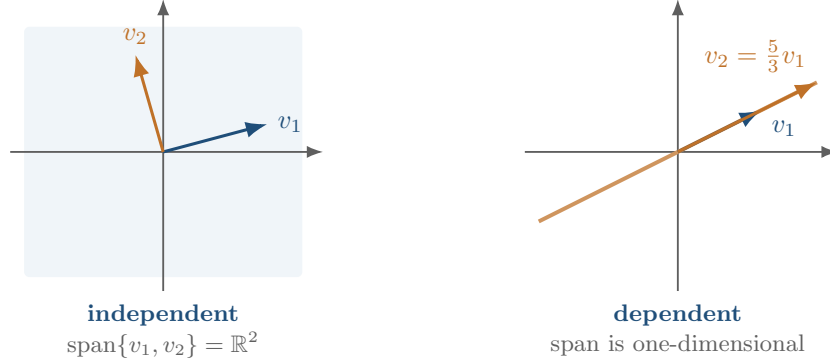


Figure 4: Independence is about genuinely new directions, not merely different vectors.

Column space, null space, and rank

For a matrix $A \in \mathbb{R}^{m \times n}$, the *column space* is $\text{Col}(A) = \{Ax : x \in \mathbb{R}^n\}$. There are at least two ways to think about what it is: 1) since Ax is a linear combination of the columns of A , the column space is exactly the span of the columns of A (hence the name), and 2) it is also the image (= the range) of the linear transformation that A represents. The *rank* of A is $\text{rank}(A) = \dim \text{Col}(A)$, so it counts the number of independent directions in the output. The *row space* of A is the span of its rows. Its dimension is called the *row rank*, while the dimension of the column space is called the *column rank*. A fundamental result is that row rank and column rank are always equal, so both are simply called the *rank*. For every $A \in \mathbb{R}^{m \times n}$, we have $0 \leq \text{rank}(A) \leq \min\{m, n\}$. This follows because A has only n columns and because $\text{Col}(A)$ is a subspace of $\mathbb{R}^m$.

The *null space* of A is $\text{Null}(A) = \{x \in \mathbb{R}^n : Ax = \mathbf{0}\}$, and it contains precisely the input directions that the transformation sends to zero. The rank–nullity theorem states that $\text{rank}(A) + \dim \text{Null}(A) = n$. Thus, the n dimensions of the input space are divided between directions that survive independently in the output and directions that are erased by the transformation. For example, consider

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

The third column is the sum of the first two, while the first two columns are independent, so $\text{rank}(A) = 2$. The same dependence appears among the rows, since the third row is the sum of the first two. Moreover, $A \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = \mathbf{0}$, and therefore $\text{Null}(A) = \text{span}\{(-1, -1, 1)'\}$. The null space has dimension 1, so rank–nullity becomes $2 + 1 = 3$ in this example. For a rectangular matrix, the phrase *full rank* must always be interpreted relative to the dimensions of the matrix. Full column rank means that the columns are linearly independent, whereas full row rank means that the rows are linearly independent. These two conditions can coincide only when the matrix is square, and for a square matrix they are equivalent to invertibility.

We have many equivalent ways to state the fact that a matrix is invertible. If $A \in \mathbb{R}^{n \times n}$, then the following statements are equivalent: A^{-1} exists; $Ax = \mathbf{0}$ has only the solution $x = \mathbf{0}$; for every $b \in \mathbb{R}^n$, the equation $Ax = b$ has exactly one solution; the columns of A are linearly independent; the columns of A span $\mathbb{R}^n$; $\text{rank}(A) = n$; and $\det(A) \neq 0$. These are not separate coincidences, because each statement expresses, in a different language, that A loses no input direction and reaches every direction in the output space.

Trace

For $A \in \mathbb{R}^{n \times n}$, the trace is the sum of its diagonal entries. The *diagonal* of a matrix always refers to the numbers going from the top-left to the bottom-right.

$$\text{tr}(A) = \sum_{i=1}^n a_{ii}.$$

For example,

$$\text{tr} \begin{bmatrix} 2 & 7 \\ -1 & 5 \end{bmatrix} = 7.$$

The trace is linear: $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$ and $\text{tr}(cA) = c \text{tr}(A)$. A particularly useful additional property is cyclicity: if $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$, then

$$\text{tr}(AB) = \text{tr}(BA).$$

More generally, factors inside a trace may be moved cyclically:

$$\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB),$$

provided the products are defined and square. They may *not* be rearranged arbitrarily; usually $\text{tr}(ABC) \neq \text{tr}(ACB)$. The thing to remember is that whatever is at the ‘back’ can be moved to the ‘front’. Why does the two-factor rule work? Expanding both sides gives the same double sum:

$$\text{tr}(AB) = \sum_i \sum_j a_{ij} b_{ji} = \sum_j \sum_i b_{ji} a_{ij} = \text{tr}(BA).$$

Applying the trace is a useful trick in econometrics as it can let us re-arrange the order of matrices. For example, since $x'Ax$ is a scalar when $x \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$, it is equal to its own trace, and so

$$x'Ax = \text{tr}(x'Ax) = \text{tr}(Axx').$$

Sometimes this kind of algebra can be exactly what you need to simplify an expression.

Slightly more advanced concepts

The concepts here are not more difficult than those in the previous section, but rely on previous concepts in order for them to be understood.

Positive definiteness

Consider first a scalar real number a . A slightly weird and convoluted definition of what it means for a to be positive is that $ax^2 > 0$ for every non-zero real number x . It turns out that the best way to generalize the concept of ‘positivity’ to matrices is by generalizing this definition; the matrix analogue of ax^2 is the *quadratic form* $x'Ax$. Throughout this section, $A \in \mathbb{R}^{n \times n}$ is real and *symmetric*⁴, meaning that $A' = A$. We say that A is

- *positive definite* (pd) if $x'Ax > 0$ for every $x \neq \mathbf{0}$;
- *positive semidefinite* (psd) if $x'Ax \geq 0$ for every x ;
- *negative definite* if $x'Ax < 0$ for every $x \neq \mathbf{0}$, and similarly we define negative semidefinite;
- *indefinite* if none of the above things are true.

Here are three concrete cases:

$$\begin{aligned} A &= \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, & x'Ax &= 2x_1^2 - 2x_1x_2 + 2x_2^2 = (x_1 - x_2)^2 + x_1^2 + x_2^2, \\ B &= \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, & x'Bx &= (x_1 + x_2)^2, \\ C &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, & x'Cx &= x_1^2 - x_2^2. \end{aligned}$$

⁴The symmetry convention is important. For any real A , $x'Ax = x'((A + A')/2)x$, because the skew-symmetric part $(A - A')/2$, whose transpose is its negative, contributes zero. Thus the quadratic form only sees the symmetric part of a non-symmetric matrix.

The first matrix is pd; the second is psd but not pd, since $x'Bx$ is zero whenever $x_1 = -x_2$; and the third is indefinite. It's important to stress that this is the correct way to define the matrix analogy to a number being 'positive', rather than something like all of its entries being positive. The matrix $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ has positive entries, but when $x = (1, -1)'$ its quadratic form equals -2 , so the matrix is indefinite. The reason why the 'definiteness' formulation is better is because many properties which are true for positive scalars are true for positive definite matrices: for example, as we will see below, the second order conditions generalize from positiveness of the second derivative in the scalar calculus case to positive definiteness of the second derivative matrix in the vector case.

Figure 5 shows level sets (= *contours*): for any given c , these are the sets $\{x : x'Ax = c\}$. In two dimensions, as in the figure, a positive definite matrix has level sets which look like ellipses that are centered at the origin. By contrast, nonsingular indefinite matrices can have hyperbolic contours that falls in some directions while rising in others. You may see someone drawing the contours of a quadratic form when doing hypothesis testing, such as when looking at the χ^2 test, as these give the shape of the region where the test is not rejected.

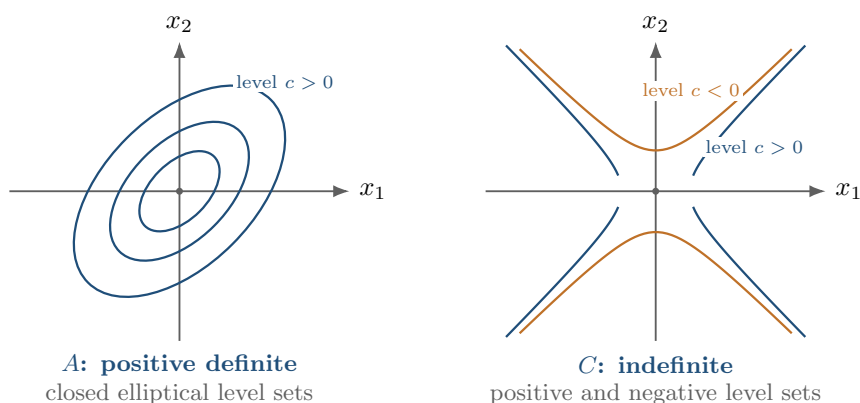


Figure 5: Level sets for the positive definite matrix A and indefinite matrix C above.

Positive definite matrices are invertible, whereas positive semidefinite matrices need not be, as B shows. This shouldn't be surprising: since inversion is the 'matrix version' of dividing, this is like say that you can always divide by a number which is strictly greater than 0, but can't necessarily divide by a number which you know is greater than or equal to 0. For the same reason, negative definite matrices are always invertible, while negative semidefinite ones need not be. A key example of a psd matrix is $X'X$ for any matrix X , since

$$x'X'Xx = (Xx)'(Xx) = \|Xx\|^2 \geq 0.$$

$X'X$ is pd exactly when X has full column rank: then $Xx \neq \mathbf{0}$ for every $x \neq \mathbf{0}$. This is the precise reason that the expression $(X'X)^{-1}$ - which appears in the formula for OLS - exists when the columns of X are linearly independent⁵. For a real symmetric matrix, positive definiteness is also equivalent to all eigenvalues being positive. We will see why after introducing diagonalization.

Projections and idempotency

Suppose $a, b \in \mathbb{R}^n$ with $b \neq \mathbf{0}$, and suppose we want to get as close as we can to a using only vectors on the line $L = \text{span}\{b\}$. The closest point on that line is the orthogonal projection of a onto L :

$$\text{proj}_b(a) = \frac{b'a}{b'b} b. \quad (4)$$

⁵When X represents the usual data matrix, then we give this the jargon of *no perfect multicollinearity*, which is one of the Gauss-Markov assumptions.

The formula follows from two requirements: the projection must lie on the line, so it has the form cb , and the residual $a - cb$ must be perpendicular to the line, so $b'(a - cb) = 0$. Solving this scalar equation gives $c = (b'a)/(b'b)$. For example, let

$$a = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Then $b'a = 5$ and $b'b = 5$, so the projection is $\hat{a} = b = (2, 1)'$. The residual is $a - \hat{a} = (-1, 2)'$, and

$$b'(a - \hat{a}) = 2(-1) + 1(2) = 0.$$

The geometry appears in Figure 6.

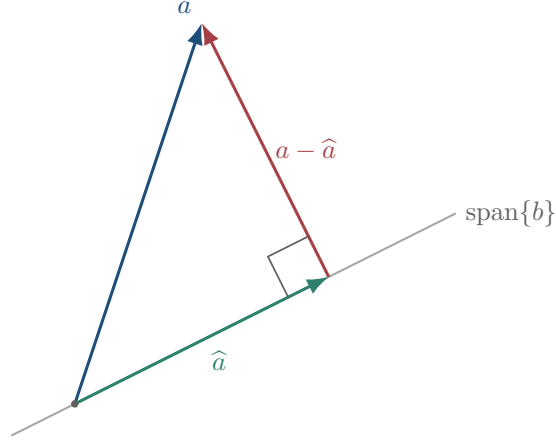


Figure 6: Orthogonal projection of a onto the line spanned by b . The residual is perpendicular to the line.

For a subspace $S \subseteq \mathbb{R}^n$, its *orthogonal complement* is

$$S^\perp = \{x \in \mathbb{R}^n : x's = 0 \text{ for every } s \in S\}.$$

It contains all vectors perpendicular to the entire subspace S .

Once again, *orthogonal* here means ‘at a right angle’: hopefully from the diagram it should make sense why going down at a right angle rather than another angle makes $a - \hat{a}$ as small as possible. On the other hand, a *projection* is any function for which doing it once is the same as doing it many times. Since a matrix $P \in \mathbb{R}^{n \times n}$ represents a linear transformation, we say that it is a *projection* if

$$PP = P.$$

This property is called *idempotency* (‘idem-potent’ is from the Latin for ‘the same - power’). It’s the right definition, as once x has been transformed by P , projecting again does nothing, since $P(Px) = Px$. By induction, we can prove that therefore it is the same as doing it many times. One important thing to note here is that an idempotent map doesn’t have to be an identity - it can do *something* - but the identity matrix is a classic example of an idempotent one. We finally have an important decomposition we can give. Note firstly that:

$$x = Px + (I - P)x.$$

The first term lies in $\text{Col}(P)$, while $P(I - P)x = (P - PP)x = \mathbf{0}$, so the second lies in $\text{Null}(P)$. This is an algebraic projection.

P will also represent an *orthogonal* projection when P is also symmetric. This is a bit more tricky to see at first glance, but comes from the fact that since $\hat{a} = Pa$ and $a - \hat{a} = (I - P)a$, we have that $\hat{a}$ and $a - \hat{a}$ are orthogonal whenever $0 = [(I - P)a]'(Pa) = a'(I' - P')Pa = a'(P - P'P)a$. A sufficient condition is that $P' = P$, as the expression is then zero by the idempotency condition $PP = P$. In the special case above where we are projecting onto the line $\text{span}\{b\}$, equation (4) can be written

$$\hat{a} = P_b a, \quad P_b = \frac{bb'}{b'b}.$$

Firstly note (by checking dimesnions) that P_b is a matrix, as $b'b$ is a scalar and bb' is a matrix. The matrix P_b is symmetric and idempotent.

Least squares as a projection

We can use the above ideas to give us another view of OLS, as this can be thought of as a projection.

The same idea explains ordinary least squares (OLS). Let $X \in \mathbb{R}^{n \times k}$ have full column rank, and suppose that we want the vector in $\text{Col}(X)$ closest to an observed vector $y \in \mathbb{R}^n$. Every vector in $\text{Col}(X)$ has the form Xb , so OLS chooses

$$\hat{\beta} = \arg \min_{b \in \mathbb{R}^k} \|y - Xb\|^2,$$

which is the usual formula for ‘minimizing the sum of squared residuals’. The notation $\arg \min$ means ‘the value of b that minimizes the expression.’ The squared norm is exactly the sum of the squared residuals:

$$\|y - Xb\|^2 = \sum_{i=1}^n (y_i - x_i'b)^2.$$

We can use this geometry intuition to come up with the formula for OLS without using any calculus. Minimizing this sum therefore means finding the point Xb in $\text{Col}(X)$ closest to y , which is achieved by projecting y orthogonally onto $\text{Col}(X)$.

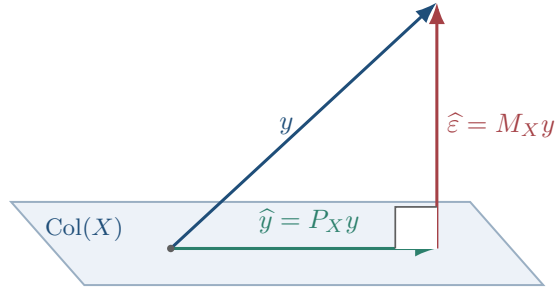


Figure 7: The least-squares fitted vector is the closest point to y in the column space of X . The residual is orthogonal to that entire space.

The residual at the closest point must meet the column space at a right angle: otherwise, moving the fitted point within the column space in the direction of the residual’s component there would shorten the residual. Hence $\hat{\varepsilon} = y - X\hat{\beta}$ is orthogonal to every column of X . Since the i -th column vector of X is given by Xe_i , where e_i has 1 in position i and 0 elsewhere, this is the same as saying that $(Xe_i)' \hat{\varepsilon} = e_i' X' \hat{\varepsilon} = 0$. Note that we can think of the identity matrix as having its i -th row as being made up of by e_i' . Therefore stacking the expressions $e_i' X' \hat{\varepsilon} = 0$ for all i gives:

$$\begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} X' \hat{\varepsilon} = \mathbf{0} \quad \text{which implies that} \quad X'(y - X\hat{\beta}) = \mathbf{0}.$$

This system of equations is known as the *normal equations* for OLS. We can then solve this (and use the fact that $X'X$ is invertible) to obtain the OLS formula.

$$\hat{\beta} = (X'X)^{-1} X'y. \tag{5}$$

The fitted values and residuals are therefore

$$\begin{aligned} \hat{y} &= X\hat{\beta} = P_X y, & P_X &= X(X'X)^{-1} X', \\ \hat{\varepsilon} &= y - \hat{y} = M_X y, & M_X &= I_n - P_X. \end{aligned}$$

Direct calculation shows

$$P'_X = P_X, \quad P_X^2 = P_X, \quad M'_X = M_X, \quad M_X^2 = M_X, \quad P_X M_X = 0.$$

Thus P_X orthogonally projects vectors onto $\text{Col}(X)$, and M_X orthogonally projects vectors onto $\text{Col}(X)^\perp$. These matrices are important in econometrics - for example, they come up in the proof of the Frisch-Waugh-Lovell theorem. By looking at the diagram, hopefully it makes sense to you why each one is symmetric and idempotent (because they represent orthogonal projections), and $P_X M_X = 0$ (because for any x, y , $P_X x$ is orthogonal to $M_X y$, and so $x'(P_X M_X)y$ always equals 0. This is only true if $P_X M_X = 0$). It's worth doing the algebra to confirm that all these properties hold too.

For full-column-rank X , the matrix

$$X^+ = (X'X)^{-1}X'$$

is known as the Moore–Penrose pseudoinverse of X , and $\hat{\beta} = X^+y$. In this full-column-rank case, X^+ generalizes an inverse to the rectangular matrix X : it satisfies $X^+X = I_k$. It's sometimes called a *left-inverse* because of this property, but note that it's not a proper inverse: when $k < n$, $XX^+ = P_X$ rather than I_n . If X is invertible, then X^+ collapses to the usual inverse. However, remember that in order for a matrix to be invertible it must be square. As such, it is definitely wrong to write $\hat{\beta} = X^{-1}y$, as the data matrix in a regression X is $n \times k$ with $k < n$. If in doubt, always make sure you double check the dimensions of things so you're not making a mistake!

Diagonalization, eigenvalues, and eigenvectors

To get to the intuition of diagonalization, we first have to remind ourselves of a way of viewing vectors. Trivially, $x = I_n x$ where I_n is the identity matrix. Letting $e_1, \dots, e_n$ be the standard basis vectors above, we note that the i -th column of I_n is given by e_i . As such, we can write any vector x as a linear combination of these basis vectors:

$$x = I_n x = x_1 e_1 + x_2 e_2 + \dots + x_n e_n$$

It's worth double checking you have got your heads around dimensions here: x and $e_1, \dots, e_n$ are all vectors, while $x_1, \dots, x_n$ are the scalar which make up the components of x . So far, this may feel like a very arbitrary thing to do. However, it has an important interpretation: it is what allows us to interpret x_1 as the co-ordinate of x in the e_1 direction, x_2 as the co-ordinate in the e_2 direction, etc. The vectors $e_1, \dots, e_n$ make up a co-ordinate system for the vector space, and we can represent x in that co-ordinate system.

However, this is not the only co-ordinate system we could use. All we need for something to work as a valid co-ordinate system for a space is for it to be a *basis*, which we defined earlier. In fact, a basis was defined exactly to have the two properties which make it valid as a co-ordinate system: linear combinations of its vectors can reach any point in the space (spanning), and this can be done in a unique way (linear independence). As such, if we have a set of vectors $p_1, \dots, p_n$ which spans $\mathbb{R}^n$, then we may be able to write the same vector x in these new co-ordinates as:

$$x = z_1 p_1 + z_2 p_2 + \dots + z_n p_n$$

$z_1, \dots, z_n$ are scalars which give the new co-ordinates for x when we use this new basis. What we have here is a different *representation* of x ; the vector x is the same, but we have two different ways of writing it.

$$p_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad p_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

The vector $x = (4, 2)'$ is $2p_1 + 2p_2$, so its coordinates in this basis are $z = (2, 2)'$. Indeed, $P^{-1}x = (2, 2)'$. Figure 8 depicts the same vector using both coordinate systems.

Note that we can write this above equation in matrix form as $x = Pz$, where P is the matrix whose i -th column is p_i , and z is a vector whose i -th component is z_i . Finally, recall from above that if the vectors $p_1, \dots, p_n$ are a basis, then P must be square and have full rank, so P is invertible. As such, we can recover what the co-ordinates of x would be in the new system by calculating $z = P^{-1}x$. This framework above may seem somewhat arbitrary, but it is important to underline what it allows us to do: we can start with a vector

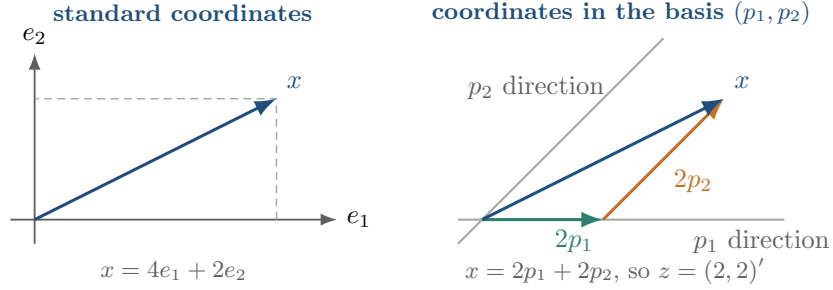


Figure 8: The same geometric vector in the standard basis and in the basis formed by the columns of P .

x , and then interpret $P^{-1}x$ as the same vector written *in a new co-ordinate system*, given by the columns of P .

For a square matrix A , a non-zero vector v is an *eigenvector* if

$$Av = \lambda v$$

for some scalar λ . The scalar λ is the *eigenvalue* associated with v .

Finally, to motivate diagonalization, we need to recall that matrices represent linear transformations. As discussed, when a linear transformation is applied to the grid, it keeps the origin fixed and maps straight lines to straight lines or single points, rather than compressing or stretching different parts of the grid by different amounts. This can actually encompass quite a large range of complicated transformations. The idea of diagonalization is the following: instead of looking at the transformation in one go, we break it into three steps: 1) transform the vector into a new system of co-ordinates, 2) apply a ‘simple’ transformation in those new co-ordinates, and then 3) transfer back to the original co-ordinates. It turns out that we can phrase many different matrices in these way⁶. We do so by writing A as the product of three matrices, which represent these three steps. More specifically, the ‘simple’ transformation in the middle is a diagonal matrix; a matrix which only has numbers on its (top-left to bottom-right) diagonal, and zeros elsewhere

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}, \quad \text{which is sometimes written as } \text{diag}(\lambda_1, \dots, \lambda_n)$$

The reason why this represents a ‘simple’ transformation is that it acts co-ordinate by co-ordinate. Convince yourself that

$$Dz = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} = \begin{bmatrix} \lambda_1 z_1 \\ \lambda_2 z_2 \\ \vdots \\ \lambda_n z_n \end{bmatrix}.$$

We say A is *diagonalizable* if there is an invertible P and a diagonal matrix D such that

$$A = PDP^{-1},$$

for a diagonal D . This factorization is called an *eigendecomposition*. The columns of P are eigenvectors of A , and the corresponding diagonal elements of D are their eigenvalues. As such, knowing the eigenvalues and eigenvectors is enough information to reconstruct A . Remember we view the product of matrices from right to left, so this represents moving to the co-ordinates defined by the columns of P (by P^{-1}), then applying the simple transformation (D), then moving back to the original co-ordinates (P).

⁶If we allow for complex numbers, then many additional matrices can be phrased in this way. However, we stick to real numbers for now.

Vectors $p_1, \dots, p_n$ are *orthonormal* if each has length one and $p_i'p_j = 0$ whenever $i \neq j$. A square matrix P is *orthogonal* if its columns are orthonormal; equivalently, $P'P = PP' = I$, or $P^{-1} = P'$. A matrix is *orthogonally diagonalizable* if it has an eigendecomposition in which P is orthogonal. The important result here is that real-valued symmetric matrices are always orthogonally diagonalizable, which comes up in econometrics all the time - and means that you often don't have to manually check if a matrix can be diagonalized.

To compute the eigenvalues and eigenvectors, we use the equation $Av = \lambda v$. For a worked example, let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}.$$

To find the eigenpairs, note that $Av = \lambda v$ is equivalent to $(A - \lambda I)v = \mathbf{0}$. A non-zero solution exists only when $A - \lambda I$ is singular, so we first solve the *characteristic equation*

$$\det(A - \lambda I) = \det \begin{bmatrix} 2 - \lambda & 1 \\ 0 & 3 - \lambda \end{bmatrix} = (2 - \lambda)(3 - \lambda) = 0.$$

Thus $\lambda = 2$ or $\lambda = 3$. Solving $(A - 2I)v = \mathbf{0}$ gives vectors proportional to $(1, 0)'$, while solving $(A - 3I)v = \mathbf{0}$ gives vectors proportional to $(1, 1)'$. Choose $v_1 = (1, 0)'$ and $v_2 = (1, 1)'$; they satisfy $Av_1 = 2v_1$ and $Av_2 = 3v_2$. Thus

$$P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix},$$

and multiplication verifies $A = PDP^{-1}$. Figure 9 shows why these basis vectors are special.

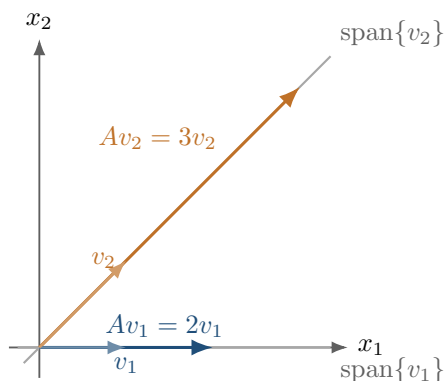


Figure 9: Eigenvectors stay on their original lines; only their scale changes.

An $n \times n$ matrix is diagonalizable over $\mathbb{R}$ if and only if it has n linearly independent real eigenvectors (as of course if it didn't, then they would not form a basis of $\mathbb{R}^n$, and so a matrix P made out of them would not be invertible). Not every matrix is diagonalizable; for example,

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

has only one independent eigendirection and is not diagonalizable. A rotation of $\mathbb{R}^2$ through an angle other than 0 or π (modulo 2π) has no real eigenvectors at all, because it sends every real direction to a different direction; it can be analyzed using complex eigenvalues, but not diagonalized over $\mathbb{R}$.

Why diagonalization is useful

As discussed above, the point of using eigenvectors and eigenvalues is to write a matrix A in terms of 'a simpler transformation, which we apply in a new co-ordinate system'. There are a couple of reasons why

we want to do this. To start with, this is the easiest way to compute what it would look like to multiply a matrix by itself r times - something we denote as A^r . This is because the middle factors cancel:

$$A^r = (PDP^{-1})^r = PD^r P^{-1}.$$

Furthermore we can use the eigendecomposition to compute inverses: if no eigenvalue is zero, the inverse is $A^{-1} = PD^{-1}P^{-1}$; it's easy to do the algebra yourself to see that $[PDP^{-1}][PD^{-1}P^{-1}] = I$, and the same is true if things are the other way round.

As mentioned above, real symmetric matrices have particularly useful structure. The spectral theorem says that every real symmetric A can be written

$$A = Q\Lambda Q',$$

where the columns of Q are orthonormal eigenvectors, $Q'Q = QQ' = I$, and Λ is real and diagonal. Orthonormal means that the columns are mutually perpendicular unit vectors. As a result⁷,

$$\text{tr}(A) = \sum_{j=1}^n \lambda_j, \quad \det(A) = \prod_{j=1}^n \lambda_j,$$

and, for symmetric A , the rank is the number of non-zero eigenvalues. The spectral theorem also explains positive definiteness. If $z = Q'x$, then

$$x'Ax = x'Q\Lambda Q'x = z'\Lambda z = \sum_{j=1}^n \lambda_j z_j^2.$$

This is positive for every non-zero x exactly when every eigenvalue is positive, and non-negative for every x exactly when every eigenvalue is non-negative.

In a sense, the eigenvalues of a matrix represent its ‘essential features’ of a matrix, and provide the most natural way to generalize concepts from scalars to matrices. For example, we know that we can divide by a number whenever it is non-zero; similarly, a square diagonalizable matrix is invertible if and only if it has no eigenvalues equal to 0. We can phrase many classic concepts in linear algebra in terms of these eigenvectors and eigenvalues, so they give us a helpful new lens on the math. Rest assured too that you won't ever have to calculate these for yourself: efficient computer algorithms can easily calculate these for any matrix you choose.

Calculus in matrix notation

One topic which intimidates a lot of students when they first see it is the idea of doing calculus in matrix notation: for example, when they see an object such as $\partial f / \partial x$, when x is a vector. The important thing to know about this is that what we are doing here is nothing new compared to ordinary multivariate calculus. All we are doing is defining a new notation for familiar objects. For a differentiable function $f : \mathbb{R}^k \rightarrow \mathbb{R}$, $\nabla_x f(x)$ is the gradient of f with respect to x : the vector of partial derivatives. We use the convention that this vector is a column, so

$$\nabla_x f(x) = \frac{\partial f(x)}{\partial x} = \begin{bmatrix} \partial f / \partial x_1 \\ \vdots \\ \partial f / \partial x_k \end{bmatrix} \in \mathbb{R}^k.$$

If f is twice differentiable, its Hessian is the $k \times k$ matrix

$$H_f(x) = \nabla_x^2 f(x), \quad [H_f(x)]_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}.$$

Derivative-layout conventions vary between authors. In econometrics it's often standard for this to be a column, so this is what we'll use here⁸.

⁷The trace result is very easy to prove for yourself: $\text{tr}(A) = \text{tr}(PDP^{-1}) = \text{tr}(DP^{-1}P) = \text{tr}(D) = \sum_i \lambda_i$. Checking that all of these steps make sense to you is a good barometer to see how well you have understood things so far.

⁸Though in fields such as vector calculus, it's often written as a row vector instead. Of course, the only difference between these is taking transposes.

Linear and quadratic functions

We'll start with a simple example. Let $a \in \mathbb{R}^k$ be constant and define $f(x) = a'x = \sum_i a_i x_i = a_1 x_1 + a_2 x_2 + \dots + a_k x_k$. It's easy to see that $\partial f(x)/\partial x_1 = a_1$, $\partial f(x)/\partial x_2 = a_2$, and so on; more generally for any i , $\partial f(x)/\partial x_i = a_i$. We can stack these partial derivatives together to get the expression for the gradient:

$$\nabla_x f(x) = \nabla_x(a'x) = \begin{bmatrix} \partial f/\partial x_1 \\ \vdots \\ \partial f/\partial x_k \end{bmatrix} = \begin{bmatrix} a_1 \\ \vdots \\ a_k \end{bmatrix} = a.$$

As such, it makes sense to write that $\nabla_x(a'x) = a$. Again: it is important to note that we are not doing anything different to usual multivariate calculus here; instead, we are just using new notation for familiar ideas.

We can similarly come up with more 'rules' for calculus in this matrix notation. If $A \in \mathbb{R}^{k \times k}$ is constant, expanding $x'Ax = \sum_i \sum_j a_{ij} x_i x_j$ gives

$$\nabla_x(x'Ax) = (A + A')x, \quad \nabla_x^2(x'Ax) = A + A'.$$

When A is symmetric, these become $2Ax$ and $2A$. Taking $A = I$ gives the familiar special case

$$\nabla_x(x'x) = 2x.$$

As such, instead of converting an expression written in matrices into a scalar form before doing calculus, we can do calculus straight away in matrix notation. For example, suppose we want to choose x to minimize the quadratic⁹

$$f(x) = \frac{1}{2}x'Ax - b'x + c,$$

where $A \in \mathbb{R}^{k \times k}$, $b \in \mathbb{R}^k$, and $c \in \mathbb{R}$ are constant, and A is symmetric positive definite. Then

$$\nabla f(x) = Ax - b, \quad \nabla^2 f(x) = A.$$

The unique *stationary point*, at which the gradient is zero, solves $Ax = b$, so

$$x^* = A^{-1}b.$$

We can also check the second order conditions for this to be a minimum, rather than a maximum. In the one-dimensional case, a sufficient condition for a stationary point to be a strict local minimum is that the second derivative is strictly greater than 0 at that point. The matrix analogy is positive definiteness, which holds here. In fact, it is a global minimum since the second derivative is positive definite at every point.

Deriving least squares

Given this toolkit, we can now derive OLS again, this time using calculus. Define the sum of squared residuals as

$$S(b) = \|y - Xb\|^2 = (y - Xb)'(y - Xb).$$

Expanding the scalar gives

$$S(b) = y'y - 2b'X'y + b'X'Xb.$$

Therefore

$$\nabla_b S(b) = -2X'y + 2X'Xb = -2X'(y - Xb), \quad \nabla_b^2 S(b) = 2X'X.$$

If X has full column rank, $X'X$ is positive definite. Setting the gradient to zero therefore gives the unique global minimizer

$$\hat{\beta} = (X'X)^{-1}X'y,$$

which agrees with the geometric argument we gave earlier.

⁹This is known as a *quadratic* as the usual quadratic equation $ax^2 + bx + c = 0$ is the scalar special case of this

Putting the ideas together

The main concepts are different views of the same underlying structure:

- The columns of A show where the coordinate directions go; their span is $\text{Col}(A)$, the set of possible outputs.
- Rank counts the independent output directions; the null space counts the lost input directions.
- For a square matrix, no lost direction means full rank, non-zero determinant, and an inverse.
- A symmetric projection keeps the component in a chosen subspace and removes the perpendicular component. Least squares is exactly this operation.
- Eigenvectors provide directions that a transformation does not turn. When enough of them form a basis, the matrix becomes diagonal in that coordinate system.
- Positive definiteness says a quadratic form rises in every non-zero direction. It is the bridge between geometry, eigenvalues, invertibility, and optimization.

The difficulty that most people have with linear algebra is the sheer amount of terminology and different ways of viewing things. As we have discussed, there are many different ways of defining basic concepts like invertibility, all of which are equivalent to each other. The best advice I can offer you is to try to focus on the intuitions discussed above. With enough practice, you'll realize that they're all talking about the same idea, and the connections between the ideas is exactly why a fluency with linear algebra can help so much with interpretation. Please feel free to come to office hours if anything isn't clear, and best of luck!